

MasterSuite

A seven-stage mastering strip, from valve colour to a true-peak limiter.



Overview

MasterSuite puts a complete mastering chain on one panel, in seven stages: input and oversampling, valve saturation with a broad tone stack, mid / side stereo control, a vari-mu style compressor, a high-frequency limiter, a clipper and a true-peak brickwall limiter.

Every stage has its own ON switch, and switching any of them crossfades over 10 ms, so it never clicks. The saturation and clipper stages can run oversampled (2x or 4x) to keep them clean of aliasing. The plugin reports its latency to REAPER, and keeps it the same whatever you bypass, so switching a stage never shifts the timing.

OUTPUT feeds the limiter, so the CEILING always has the final say on the peak level.

Reach for it for

- Finishing a mix or a stem master in one plugin
- Gentle glue and warmth on a mix bus
- Loud, punchy masters with a clipper into the limiter
- Streaming-safe masters with a -1 dB true-peak ceiling
- Mono-ing the low end and adjusting stereo width

Controls

IN / OUT

TRIM	Input level, -12 to +12 dB.
SAT	Band saturation from the GFX output stage: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises. It sits after OUTPUT, before the limiter.
OFF / CLIP / LIMIT (OUT)	The GFX output stage, just before the true-peak limiter: nothing (default), a soft clipper at -0.3 dBFS, or a lookahead limiter at -0.3 dBFS (3 ms extra latency). The CLIP key flashes red while it is clipping.
OVERSAMPLING: OFF / 2X / 4X	Runs the TUBE and CLIPPER stages at two or four times the sample rate (2X by default) for cleaner saturation with less aliasing. Higher settings add latency.
IN meters	Input level, left and right. Each shows a floating peak; click any meter to clear the peaks.

TUBE

Asymmetric valve saturation. Small signals pass at unity; turning DRIVE up adds saturation rather than level.

ON	Switches the stage in or out (in by default).
DRIVE	How hard the valve is driven, -12 to +12 dB (-3 dB by default).
DRIFT	How much the valve's bias and drive wander, 0 to 100% (50% by default). See Drift below.
BIAS: LOW / MID / HIGH	The valve's asymmetry. LOW is the cleanest and gentlest; HIGH adds more even (2nd) harmonics and bites harder. MID is the default.

TONE

A broad four-band EQ, -6 to +6 dB on each band, with wide, gentle curves.

ON	Switches the tone stage in or out (in by default).
TAPE	Adds a gentle tape-style roll-off at 18 kHz.
BASS	A low shelf at 90 Hz.
LO-MID	A broad bell at 350 Hz - for warmth or mud.
HI-MID	A broad bell at 2.8 kHz - for presence or harshness.
PRESENCE	A high shelf at 7 kHz - for air and brightness.

STEREO

ON	Switches the stereo stage in or out (in by default).
MONO	Makes everything below this frequency mono, from OFF (fully left) up to 200 Hz. It removes the side signal below the frequency with a steep 24 dB per octave filter, for a tight, centred low end.
WIDTH	The level of the side signal, 0 to 200% (100% by default). 0% is mono; above 100% widens.

VARI-MU

A stereo-linked RMS compressor with a soft knee. Like a valve vari-mu, its ratio grows a little as it works harder, and its release slows after long stretches of gain reduction.

ON	Switches the compressor in or out (out by default).
THRESH	-40 to 0 dB (-18 dB by default).
RATIO	1:1 to 10:1 (2:1 by default).
KNEE	0 to 12 dB (6 dB by default). Wider knees ease the compression in more gradually.
MAKEUP	0 to +12 dB.

MIX	Parallel compression: 100% is fully compressed; lower blends the uncompressed signal back in.
RECTIFIER: GERM / SILICON / LED	The timing. GERM (germanium): slow and smooth (30 ms attack, 350 ms release). SILICON: medium (10 ms, 180 ms), the default. LED: fast (2 ms, 70 ms).
GR	The compressor's gain reduction, 0 to 20 dB, with the reading above it.

HF LIMIT

A fast peak limiter that listens only to the top above 6 kHz and turns down only that band, for harshness and sibilance. The split keeps both bands in phase, so at rest the response is flat.

ON	Switches it in or out (out by default).
THRESH	-30 to 0 dB (-12 dB by default). Lower catches more of the top.
GR	How far the top band is being turned down.

CLIPPER

ON	Switches the clipper in with the last mode chosen (out by default).
DRIVE	How hard the signal is pushed into the clipper at 0 dBFS, 0 to +12 dB.
SOFT / HARD	SOFT: a rounded clip that eases into the ceiling. HARD: a straight clamp. Either can be picked while the clipper is on or off.

LIMITER

A true-peak brickwall limiter: it finds the peaks between the samples, looks 1.5 ms ahead and shapes the gain so the output never goes over the ceiling.

ON	Switches the limiter in or out (in by default).
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CEILING	The maximum output level, -1.00 to -0.10 dB (-0.30 dB by default). Use -1 dB for streaming.
RELEASE	How fast the limiter recovers, 10 to 500 ms (80 ms by default).
GR	The limiter's gain reduction.

Output

OUTPUT	The level into the limiter, -12 to +12 dB. Turning it up makes the master louder and the limiter work harder; the CEILING still holds the peaks.
OUT meters	Output level, left and right.

Drift

In MasterSuite, drift makes the TUBE stage's bias and drive wander slowly, like a real valve. The DRIFT knob sits in the TUBE section: 50% is the standard amount, 100% is twice that and 0 switches it off. The wander also speeds up a little as you turn it up.

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PRESETS	Opens the list of factory presets; the arrows step through them and the display shows the one in use (click it for the list too). Choosing one sets the seven stages and OUTPUT; DRIFT and the IN / OUT strip's SAT and OUT are left alone. The last entry resets to the defaults.
Theme picker	Changes the panel's colour theme.
S / M / L	Panel size: small, medium or large.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
HW	Switches between the hardware-style panel and the flat look.

LATENCY

Shows the plugin's current latency in samples and milliseconds.

Presets

Transparent	No valve, gentle 1.5:1 glue, mono below 30 Hz, a -1 dB ceiling.
Warm Glue	Valve on, a little bass and tape roll-off, slow 2:1 compression, HF limiter and a soft clip into a -0.5 dB ceiling.
Loud & Punchy	4x oversampling, 3:1 fast compression blended at 70%, slightly wider, soft clip into a -0.3 dB ceiling.
Vintage Tape	Valve driven harder on low bias, tape roll-off, side narrowed to 90%, slow compression, a -1 dB ceiling.
Bright Pop	High valve bias, mids scooped and presence lifted, 120% width, soft clip and a -0.3 dB ceiling.
Bass Heavy	4x oversampling, bass up, mono below 150 Hz, fast compression, hard clip.
Acoustic / Classical	Almost nothing: no valve, very light 1.3:1 compression with a wide knee, slow 300 ms limiter release.
Podcast / Voice	Mono (width 0, mono below 200 Hz), bass cut and mids forward, 3:1 compression, HF limiter, a -1 dB ceiling.
Parallel Punch	Heavy 6:1 fast compression blended in at 40% under the dry mix.
Streaming Safe (-1 dBTP)	Controlled, gentle compression, no clipper, a -1 dB true-peak ceiling.
Reset to defaults	Every stage back to its starting value, including the DRIFT knob.

Tips

- Set the CEILING first, then raise OUTPUT until the limiter's GR shows only a few dB on the loudest parts.
- A little CLIPPER DRIVE in front of the limiter lets you get louder with fewer limiter artefacts; switch OVERSAMPLING to 4X when you do.
- Use HF LIMIT instead of a cut with the TONE stage when only some moments are harsh or sibilant.
- MONO at 80 - 150 Hz tightens the low end for vinyl, clubs and small speakers without narrowing the rest of the mix.
- Switch stages ON and off as you work: each one crossfades cleanly, so you can compare without clicks.

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